

WEEK 6



Inverted Pyramid

Small at the bottom. Wide on top.

거꾸로 된 피라미드 — 아래는 좁고 위가 넓게 · 영어코딩

BIG PICTURE

Flip the shape upside down 🙄



Same idea as last week — but the layers **grow**.

지난주와 반대로, 층이 점점 커집니다

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Grow instead of shrink



normal (W5)

start big $\rightarrow$ size - 2



inverted (W6)

start small $\rightarrow$ size + 2

Start at 1, add 2 each layer until you reach the top.

CONCEPT

New condition

```
size = 1
while size <= 9:
    build_layer(size)
    agent.move(UP, 1)
    size = size + 2
```

Loop while `size <= 9` — stop once it passes 9.

size가 9 이하인 동안 반복

WATCH IT!

size grows, then exits

```
size = 1
```

```
size <= 9
```

```
build_layer(size); move up; size += 2
```

```
▸ layers built
```

CODE

Two changes from last week

```
size = 1                # start small

while size <= 9:         # grow until 9
    build_layer(size)
    agent.move(UP, 1)
    size = size + 2      # add, don't subtract
```

Same build_layer — only the start, condition, and step flip.

PREDICT 🤖

Where is the widest layer?

Predict first — then click reveal

```
size = 1
while size <= 9:
    build_layer(size)
    agent.move(UP, 1)
    size = size + 2
```

Reveal answer 🔍 reveal

VOCAB

New words today

inverted flipped upside down

grow get bigger each layer (size + 2)

<= less than or equal to

widest the biggest layer

거꾸로

커지다

작거나 같다

가장 넓은

 DEBUG

What's wrong here?

buggy

```
# goal: inverted, smallest at bottom  
size = 9  
while size > 0:  
    build_layer(size)  
    agent.move(UP, 1)  
    size = size - 2
```

Think first — then click next.



DEBUG

The fix

Bug: This is the *normal* pyramid — it starts big and shrinks. For inverted, start at 1, grow with `+ 2`, and loop `while size <= 9`.

fixed

```
size = 1
while size <= 9:
    build_layer(size)
    agent.move(UP, 1)
    size = size + 2
```

YOUR TURN!

Build an inverted pyramid ▼

- ✓ Start with `size = 1`
- ✓ Loop `while size <= 9`, add 2 each layer
- ✓ Move up 1 between layers (1 → 3 → 5 → 7 → 9)
- ✓ Look from the side — the top is widest
- ✓ Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

In an inverted pyramid, where is the widest layer?

In the middle.

On the bottom.

On top.

There is no widest layer.



QUIZ

QUESTION 2 OF 3

Which step makes the layers **grow**?

size = 0

size = size - 2

size = size - 1

size = size + 2



QUIZ

QUESTION 3 OF 3

Compared to the normal pyramid, what changes?

The start value, the condition, and the step (+2 not -2).

Nothing changes.

Only the block color.

You remove the while loop.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Flipped it!

Next: hollow, stepped, and custom pyramid variations.

다음 주: 속 빈·계단식 등 나만의 변형 만들기